

ARTICLES

Tomato-Potato Late Blight in the Home Garden

This article describes tomato-potato late blight, including symptoms; disease development, cycle, and management; and management after harvest.



Introduction

The causal pathogen of late blight from potatoes was first described in 1845 by Montagne, and from tomato in 1847 by Payen in France. However, it was 1863 before deBary established beyond doubt that this organism, *Phytophthora infestans*, was the cause of late blight. The disease occurs worldwide where tomatoes and potatoes are grown.

Late blight is thought to have

originated in Central America and to have appeared almost simultaneously in Europe and North America about 1830. It occurred in France in 1840, was destructive in Germany in 1841, and appeared in North America in 1843. Then, from 1844 to 1847, it occurred in epidemic and catastrophic proportions throughout Europe and North America. It was directly responsible for the Irish Potato Famine of 1845 and 1846. Since that time, epidemics have happened periodically when weather and other conditions favored disease development.

Late blight epidemics occurred in 1878 in England, destroying entire plantings; 1906, 1927, and 1928 in California; 1940 in Ontario; 1946 and 1947 throughout the eastern half of North America when 80 to 90 percent of early seedbeds in Florida were a complete loss, over 50 percent of the crop was lost in eastern states from New York to Florida, and 25 percent of the crop was lost in midwestern states; 1960 in Ontario; and 1976 in southern Georgia and some northern states where infected transplants were used.

Heavy losses can take place in transit; symptoms can occur on infected but symptomless tomato fruit within 5 days of harvest. In addition to tomato and potato, the late blight pathogen has also been reported to infect eggplant and pepper, although no cases have been reported recently. The pathogen may infect solanaceous weeds such as bittersweet nightshade, as well as other plants in the same botanical (*Solanaceae*) family, including petunias, Chinese lantern, and tomatillos.

Similar to the seasonal flu in humans, different isolates (genotypes) of the late blight pathogen exist and infect different hosts; so it is highly unlikely that a given isolate of the pathogen would be capable of infecting and causing disease on all known host plants. The genotypes most common in Pennsylvania and the Northeast can cause disease in either potato or tomato or both. In 2009, the majority of losses for tomato and some potato losses were due to the relatively new US22 genotype. Other losses for potato were due to the USB genotype which is widespread and commonly seen on potato annually.

Late blight is not uncommon in Pennsylvania. Each year, there are several confirmed outbreaks on potato and/or tomato in commercial fields. The occurrence of late blight in 2009 was different for several reasons. One is that this was the earliest the disease had been reported over such a broad region of the country (the Northeast). By mid-June, late blight had been confirmed in several states in the Northeast, including Pennsylvania. Another reason was the distribution of infected transplants through local retail stores from Ohio to Maine. Never before had such an extensive distribution of late blight-infected plants occurred.

In addition, the cool and wet conditions that favor disease development persisted through the growing season. Keep in mind that in order for the disease to develop, the pathogen (*Phytophthora infestans*) must be present in combination with a susceptible host (primarily tomato, potato) and environmental conditions favorable for the pathogen to develop and cause disease. If any one or more of these components is absent, then late blight will not develop. In 2009, all three components were present throughout the growing season.

Symptoms

Late blight symptoms can develop on leaves, stems, branches, and both green and ripe fruit; and they are very obvious to the naked eye. On leaves, pale green to brown spots appear on the upper surfaces. Leaf spot margins often are pale green or water-soaked. The spots may enlarge rapidly until entire leaflets are killed. In moist conditions, a downy white growth usually develops near the margins of leaf spots on the undersides of leaves. This white growth contains the spores that are easily blown around in the wind. When petioles and stems are affected, portions of plants beyond blight lesions may dry up rapidly and collapse.

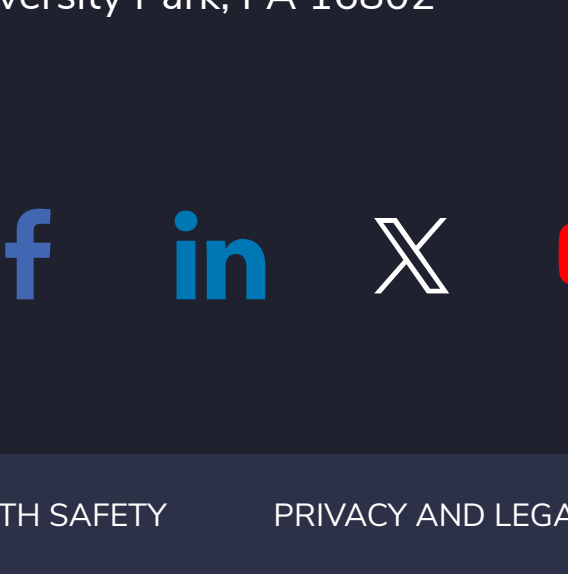
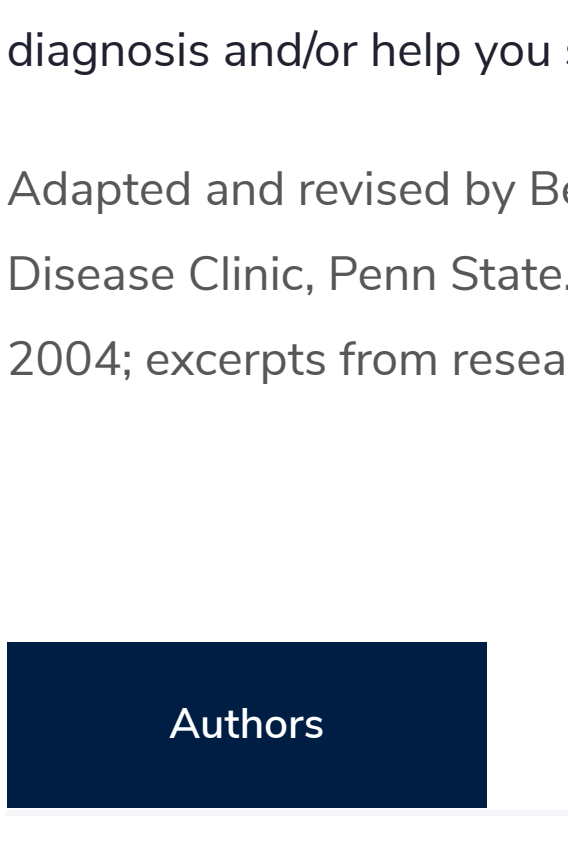
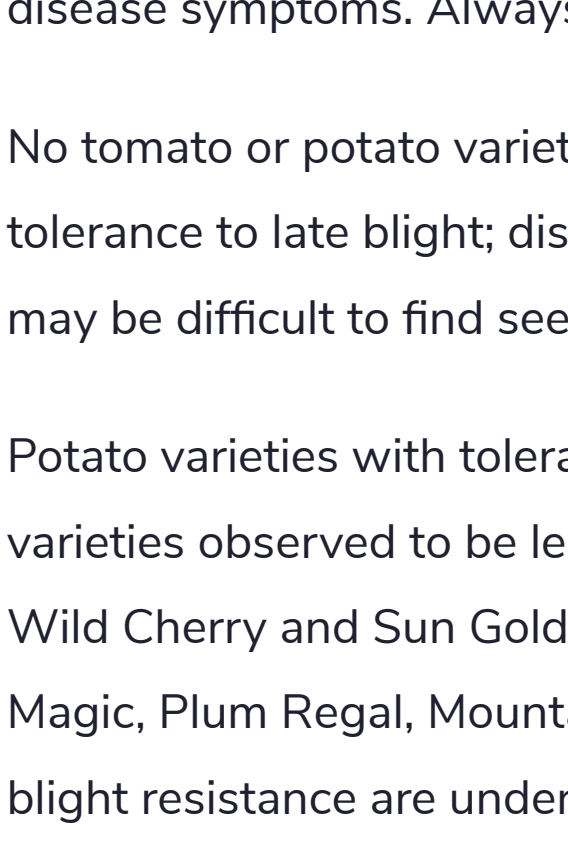
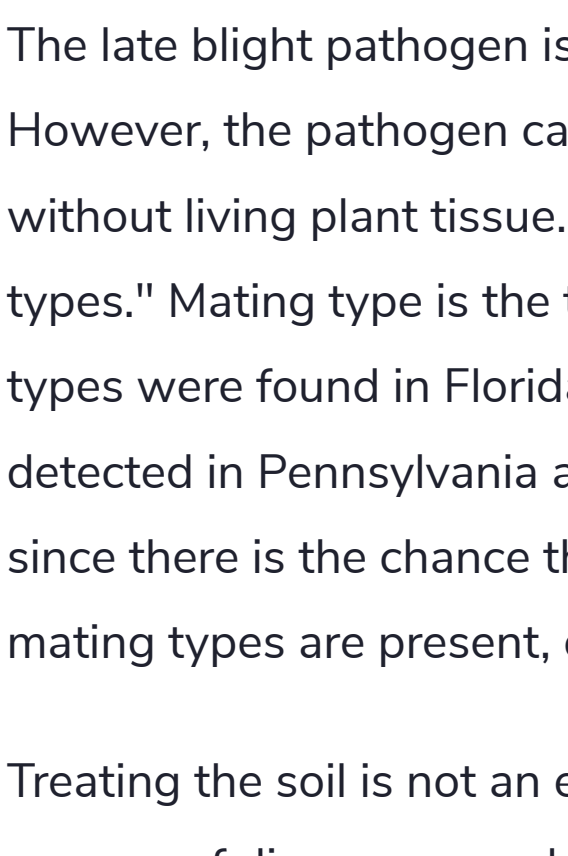
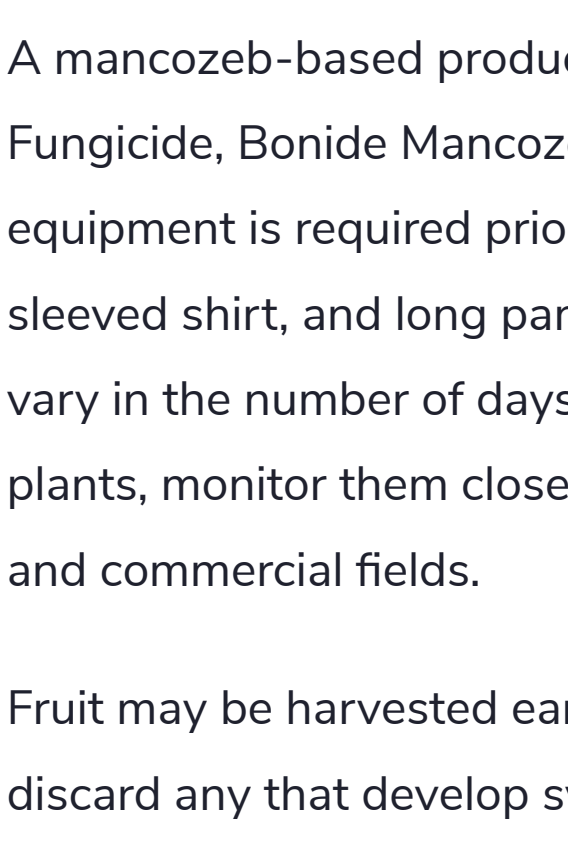
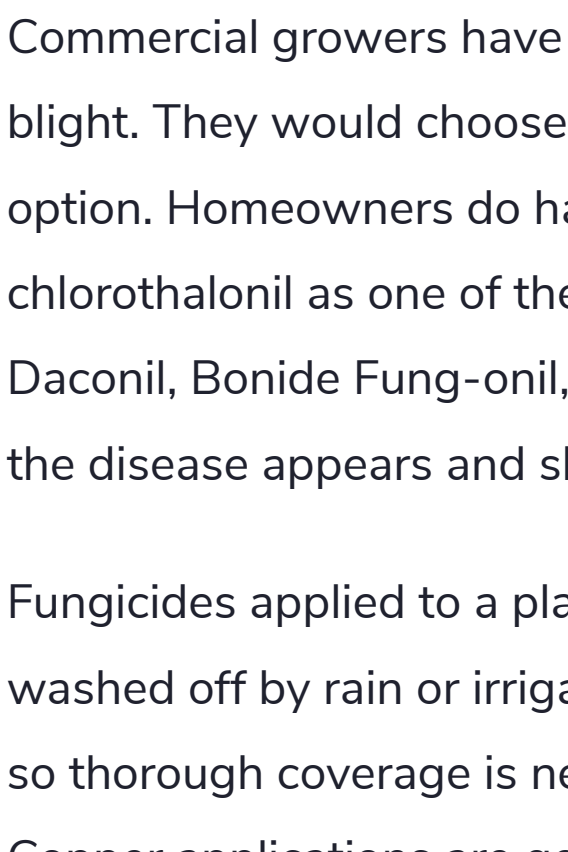
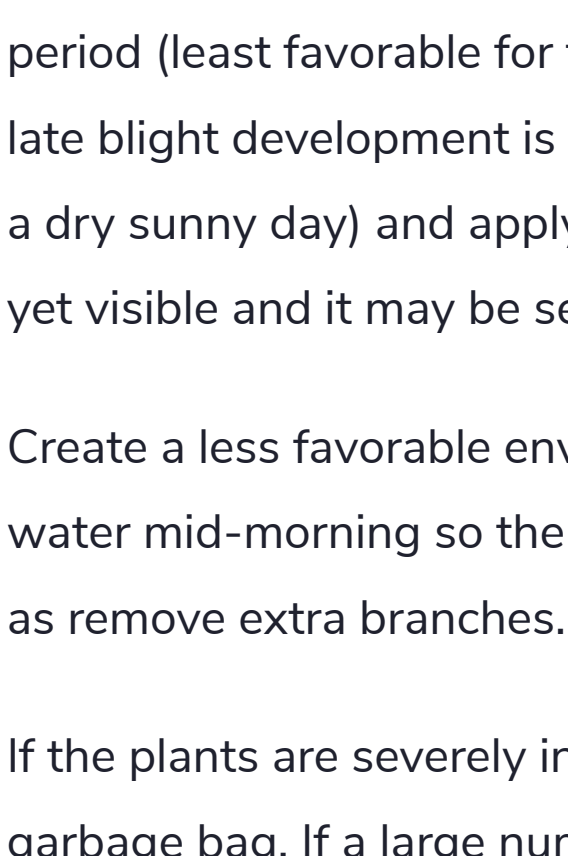
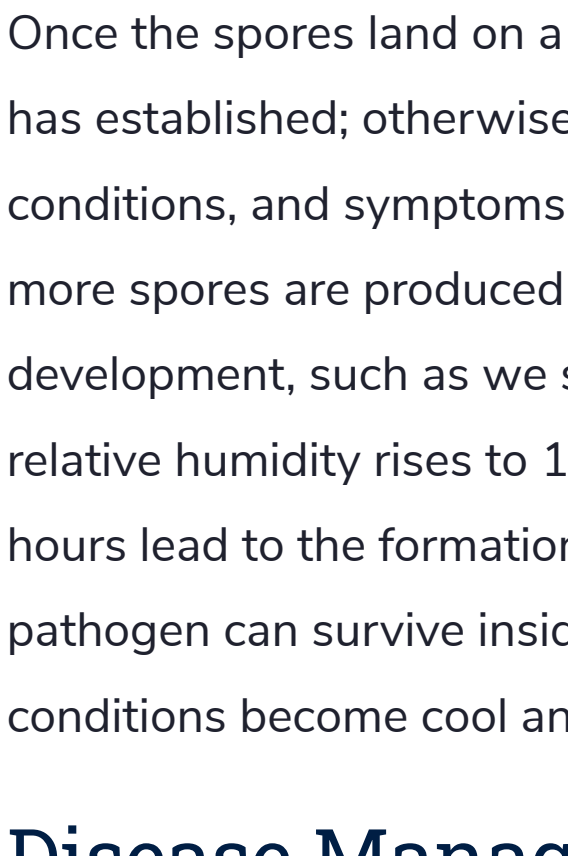
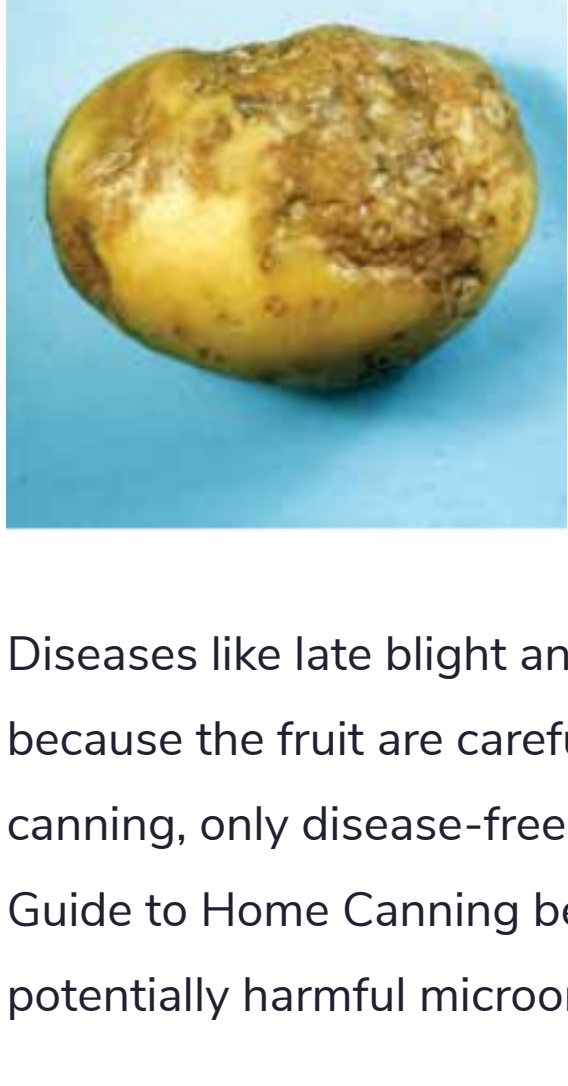
Lesions can expand rapidly and result in extensive, if not complete, defoliation within 14 days. In dry weather, affected foliar parts may appear dry and shriveled. Stem lesions are typically brown to almost black in color.

Tomato fruit can become infected when foliage is affected. On fruit, greenish-brown greasy-appearing spots may enlarge until the entire fruit is involved. The fruit tissue remains firm at first, with varying depths of discolored tissue below the skin. In moist weather, a white downy growth may appear on the affected fruit rot surface. Secondary organisms may invade affected fruit and cause a soft rot. Infected fruit can develop directly on the plant or a few days after sitting on a kitchen counter.

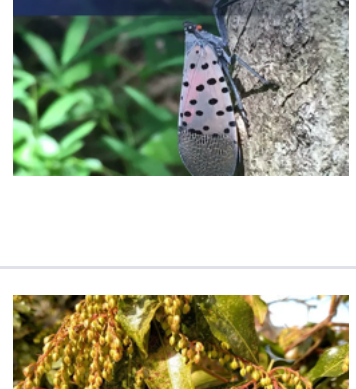
Potato tubers can become infected from spores splashing down into the soil. Late blight spots on tubers are reddish brown, dry, and granular. These spots often become infected with secondary pathogens such as bacteria, which can cause the entire tuber to rot and turn soft.

Although the unaffected parts of the tomato fruit are probably safe to eat, no published scientific study on this specific issue has been found to confirm this conclusion. Therefore, consumers need to make their own decisions about food safety. The conclusion that unaffected tissue is safe to consume is based on several points. This pathogen does not produce a toxin that can make people sick, as a few plant pathogens can do. Plant pathogens cannot infect people. No food safety issues have been found for other diseases that affect tomato fruit or potato tubers. Late blight appears to be like other more common diseases—for example, anthracnose on tomato fruit and pink rot of potato (which, incidentally, is caused by *Phytophthora erythroseptica*, a pathogen related to that causing late blight)—in that these do not appear to affect plant tissue beyond the area of infection.

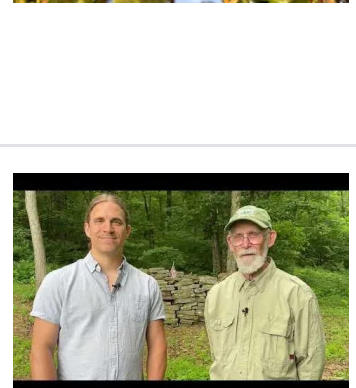
Many home gardeners often cut off diseased tissue rather than throw out the entire fruit or tuber, having found the healthy-appearing parts of these to taste fine. Keep in mind, however, that infection can create conditions in which other potentially harmful secondary organisms can also invade tissue and grow.



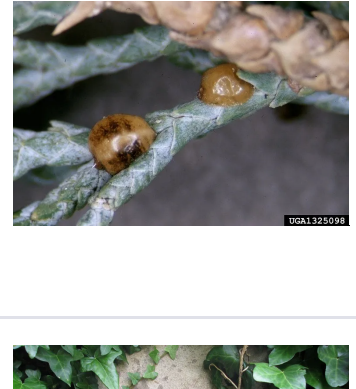
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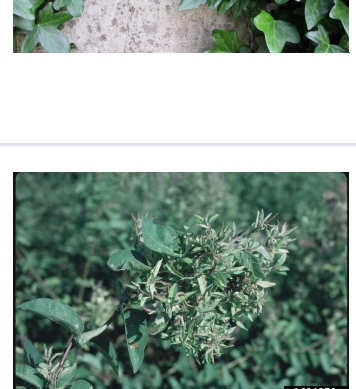
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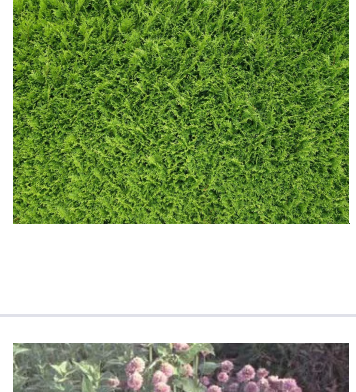
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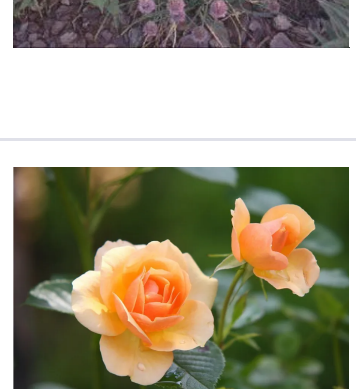
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